# EC-390: Digital Signal Processing

# *LECTURE NO 7: Classification of Discrete Time Systems*

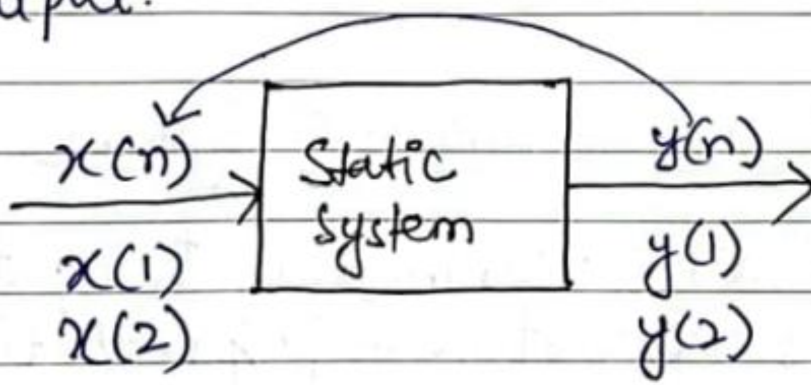
**Engr. Tehseen Ahsan**
**Assistant Professor, Computer Engineering Department**
**HITEC University Taxila Cantt, Pakistan**

# Organization of Lecture

1. Static and Dynamic Systems.
2. Time Invariant and Time Variant Systems.
3. Linear and Non-Linear Systems.
4. Causal and Non-Causal Systems.
5. Stable and Unstable Systems.

# 1. Static and Dynamic Systems

* Static Systems are also called Memory less Systems
* A D.T System is said to be static system if its output at any instant 'n' depends only on the input applied at that instant 'n' but not on the past or future values of input or past values of output.




$x(n)$ = present I/p
$x(n-1)$ = past I/p
$x(n+1)$ = Future I/p

Examples:

$y(n) = x(n) \Rightarrow$ static system

$y(n) = n x(n) + 2x^2(n) \Rightarrow$ static system

# 1. Static and Dynamic Systems

* A D.T system is said to be dynamic or memory system, if its output at any instant 'n' depends upon past or future inputs or past outputs.
* All time shifting signals are dynamic.
* Time scaling signals are also dynamic.

* Summation cases signals are also dynamic signals

Examples:

$y(n) = x(2n) \rightarrow$ Time scaling, Dynamic signal

$y(n) = x(n) + x(n-2) \rightarrow$ Time shifting, Dynamic Signal

$y(n) = \sum_{m=0}^{\infty} x(n-m) \rightarrow$ Summation + shifting signal, Dynamic signal.

$y(n) = x(n) + 4x(n+1) \rightarrow$ Time shifting, Dynamic signal.

# 1. Static and Dynamic Systems

**Home Work Examples:**

Find whether the following systems are dynamic or static

(i) $y(n) = x(n+2)$

(ii) $y(n) = 2x^3(n)$

(iii) $y(n) = x(n-4) + x(n)$

# 2. Time Invariant and Time Variant Systems

* A time invariant system means its input-output characteristics are not changing with timeshifting

$$x(n) \rightarrow \boxed{\text{System } \mathcal{H}} \rightarrow y(n) \qquad x(n-k) \rightarrow \boxed{\text{System } \mathcal{H}} \rightarrow y(n-k)$$

$$x(n) \xrightarrow{\mathcal{H}} y(n) \qquad x(n-k) \xrightarrow{\mathcal{H}} y(n-k)$$

Time Invariant System

# 2. Time Invariant and Time Variant Systems

How to determine whether the system is time invarient or not?

Step1: Delay the input $x(n)$ by 'K' units that means $x(n-K)$. Denote the corresponding output by $y(n,K)$

$$x(n-K) \xrightarrow{\mathcal{H}} y(n-K)$$

Step2: In the given equation of system $y(n)$ replace 'n' by 'n − k' throughout. This output is $y(n-k)$

Step3: If $y(n,K) = y(n-K)$ then system is time invarient and if $y(n,K) \neq y(n-K)$ then system is time varient

# 2. Time Invariant and Time Variant Systems

Example 1: Determine whether the following systems is time invariant or not.

$$y(n) = x(n) - x(n-1)$$

Sol$^n$:

Step 1: Delay the input by 'k' samples and denote corresponding output by $y(n,k)$

$$\therefore y(n,k) = x(n-k) - x(n-k-1) \quad \text{--- (i)}$$

Step 2: Replace 'n' by 'n-k' throughout the given equation and denote corresponding output by $y(n-k)$

$$\therefore y(n-k) = x(n-k) - x(n-k-1) \quad \text{--- (ii)}$$

Step 3: Compare Equations (i) and (ii)

Here $y(n,k) = y(n-k)$

Thus the system is time invariant.

# 2. Time Invariant and Time Variant Systems

Example 2: Determine whether the following systems is time invariant or not.

$$y(n) = x(n) + n\,x(n-1)$$

Sol$^n$:

Step 1:

$$y(n,k) = x(n-k) + n\,x(n-k-1) \quad \text{— (i)}$$

Step 2:

$$y(n-k) = x(n-k) + (n-k)\,x(n-k-1) \quad \text{— (ii)}$$

Step 3: Compare eq (i) & (ii)

$$\therefore\ y(n,k) \neq y(n-k)$$

Thus the system is time varient

# 2. Time Invariant and Time Variant Systems

**Home Work Examples:**

∴ Determine whether the following systems are time invariant or not.

1) $y(n) = x(n^2)$
2) $y(n) = x(2n)$
3) $y(n) = y(n-4) + x(n-4)$
4) $y(n) = x(-n)$

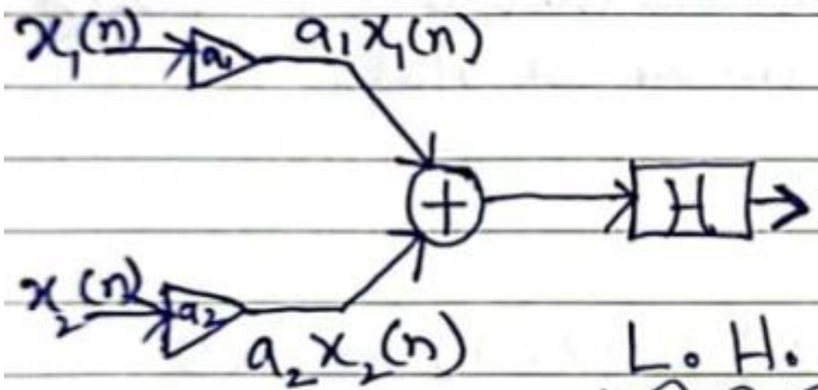
# 3. Linear and Non-Linear Systems

* A linear system is a system which follows superposition principle
* A system is said to be linear if the combined response of $a_1x_1(n)$ and $a_2x_2(n)$ is equal to the addition of individual responses.

$$\mathcal{H}[a_1x_1(n)+a_2x_2(n)] = a_1\mathcal{H}[x_1(n)]+a_2\mathcal{H}[x_2(n)] \quad -(1)$$

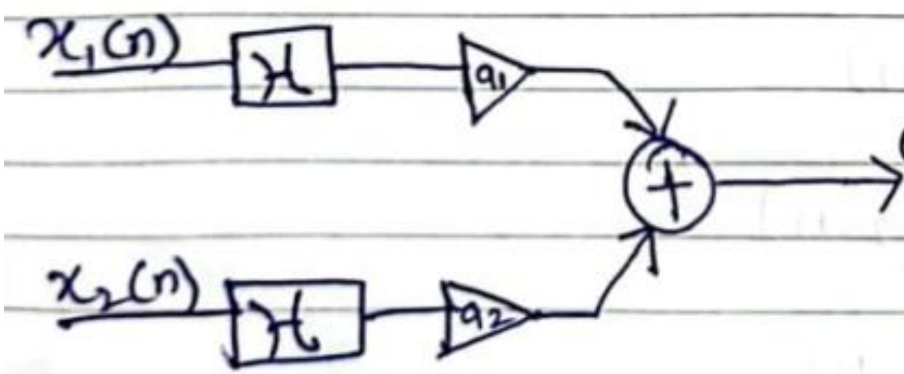
Equation above (eq (1)) is called as superposition theorem. The graphical representation of L.H.S and R.H.S of eq (1) is shown below

# 3. Linear and Non-Linear Systems




$$\mathcal{H}[a_1 x_1(n) + a_2 x_2(n)] = y'(n)$$

L.H.S of eq ①




$$a_1 \mathcal{H}[x_1(n)] + a_2 \mathcal{H}[x_2(n)] = y''(n)$$

R.H.S of eq ①

# 3. Linear and Non-Linear Systems

How to determine whether the given system is linear or Non-linear?

(i) For $x_1(n) \xrightarrow{\mathcal{H}} y_1(n)$

(ii) For $x_2(n) \xrightarrow{\mathcal{H}} y_2(n)$

(iii) $y'(n) = y_1(n) + y_2(n)$

(iv) For $y''(n)$, substitute $x(n) = x_1(n) + x_2(n)$ in your given equation.

(v) If $y'(n) = y''(n) \Rightarrow$ system is Linear.

(vi) If $y'(n) \neq y''(n) \Rightarrow$ system is non-linear.

# 3. Linear and Non-Linear Systems

Example 1: Determine whether the following system is linear or not.

$y(n) = nx(n)$.

Solⁿ:

Given equation: $y(n) = nx(n)$ —— (i)

(i) $x_1(n) \xrightarrow{\mathcal{H}} y_1(n)$

$y_1(n) = nx_1(n)$

(ii) $x_2(n) \xrightarrow{\mathcal{H}} y_2(n)$

$y_2(n) = nx_2(n)$

(iii) $y'(n) = y_1(n) + y_2(n)$

$y'(n) = nx_1(n) + nx_2(n)$

$y'(n) = n[x_1(n) + x_2(n)]$

# 3. Linear and Non-Linear Systems

(iv) For $y''(n)$, substitute $x(n) = x_1(n) + x_2(n)$
in eq. (i)

$y''(n) = n[x_1(n) + x_2(n)]$

$y'(n) = y''(n)$ ; system is Linear.

# 3. Linear and Non-Linear Systems

Example 2: Determine whether the following system is linear or not.

$$y(n) = x(n) + n x(n+1)$$

Sol$^n$:

$$y(n) = x(n) + n x(n+1) \quad \text{—— (i)}$$

(i) $y_1(n) = x_1(n) + n x_1(n+1)$

(ii) $y_2(n) = x_2(n) + n_1 x_2(n+1)$

(iii) $y'(n) = y_1(n) + y_2(n)$

$$y'(n) = x_1(n) + n x_1(n+1) + x_2(n) + n x_2(n+1)$$

$$= x_1(n) + x_2(n) + n x_1(n+1) + n x_2(n+1)$$

$$y'(n) = x_1(n) + x_2(n) + n\left[x_1(n+1) + x_2(n+1)\right]$$

# 3. Linear and Non-Linear Systems

(iv) For $y''(n)$:

- substitute $x(n) = x_1(n) + x_2(n)$ and $x(n+1) = x_1(n+1) + x_2(n+1)$ in eq (i)

$$y''(n) = x_1(n) + x_2(n) + n\left[x_1(n+1) + x_2(n+1)\right]$$

(v) Compare $y'(n)$ & $y''(n)$

$y'(n) = y''(n) \Rightarrow$ system is Linear.

Home work Examples

Determine whether the following systems are linear or not

1) $y(n) = 10x(n) + 5$
2) $y(n) = \cos x(n)$
3) $y(n) = x(2n)$
4) $y(n) = x(-n+2)$

# 4. Causal and Non-Causal Systems

* A system is said to be causal if its output at any time depends only on present input and/or past inputs. But the output does not depend on future inputs.

Examples:

(i) $y(n) = x(n) + x(n-1)$

(ii) $y(n) = 3x(n)$

(iii) $y(n) = x(n) + 4x(n-1)$

# 4. Causal and Non-Causal Systems

* A system is said to non-causal if its output depends not only on present and past inputs but also on future inputs.

Examples:

(i) $y(n) = x(n) + x(n+1)$

(ii) $y(n) = Bx(n+2)$

(iii) $y(n) = x(n) + nx(n+1)$

# 5. Stable and Unstable Systems

* To define stability of a system we will use the term BIBO. It stands for Bounded Input Bounded Output. The meaning of word 'bounded' is some finite value. So bounded input means input signal is having some finite value i.e., input signal is not infinite. Similarly bounded output means the output signal attains some finite value i.e., the output is not reaching to infinite level.

* Stable system: An initially relaxed system is BIBO stable if and only if every bounded input produces bounded output.

# 5. Stable and Unstable Systems

Here a relaxed system means when input to the system is zero then the output of system is also zero.

* Unstable system: An initially relaxed system is said to be unstable if bounded input produces unbounded (infinite) output.

* Examples of bounded input signal are step signal, decaying exponential signal and impulse signal.

* Examples of unbounded input signal are ramp signal and increasing exponential signal.

## References: Textbooks

1. *"Discrete-Time Signal Processing", International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *"Digital Signal Processing Using Matlab", 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*